

ReviewAgent

Project Status Report Complete & Incomplete Items

70 files created | 15 items remaining

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COMPLETED ITEMS

A. Documentation (6 files)

#	File	Description
1	ReviewAgent_Project_Proposal.md	Full proposal: problem statement, 5 contributions, 3 RQs, theoretical foundations, 38 references
2	ReviewAgent_Detailed_Architecture.md	Technical deep-dive: all stages with theory tags, JSON schemas, examples
3	ReviewAgent_Experimental_Design.md	3 experiments, 7 ablations, statistical tests, sample size justification
4	ReviewAgent_Research_Proposal.pdf	16-page polished PDF with all 15 sections
5	generate_pdf.py	PDF generation script using fpdf
6	IMPLEMENTATION_GUIDE.md	Beginner-friendly guide explaining every file, setup, and usage

B. Project Setup (3 files)

#	File	Description
7	pyproject.toml	30+ dependencies: torch, transformers, networkx, chromadb, trl, fastapi, etc.
8	.env.example	API key template for OpenAI, Anthropic, HuggingFace
9	.gitignore	Ignores data/, models/, venv/, __pycache__/

C. Configuration (7 files)

#	File	Description
10	configs/base.yaml	Shared: LLM provider, model name, thresholds
11	configs/stage1.yaml	RoBERTa: 7 labels, training params, confidence threshold 0.7
12	configs/stage2.yaml	KG node/edge types, HDBSCAN params, priority weights
13	configs/stage3.yaml	Translation temperature 0.3, HITL min score 3.0
14	configs/stage4b.yaml	ChromaDB path, 5 RAG sources, 3 refinement iterations
15	configs/stage5.yaml	LoRA config, KTO/DPO/PPO hyperparams, feedback dimensions
16	configs/experiments.yaml	All 3 experiments + 7 ablations fully configured

D. Common Module (3 files)

#	File	Description
17	src/common/schemas.py	8 Pydantic models: ReviewObject, AspectSentiment, ExtractedEntities, IssueCluster, IssueSpec, RubricScores, ComplianceFlags, GeneratedResponse
18	src/common/config.py	YAML config loader using OmegaConf
19	src/common/llm_client.py	Unified OpenAI + Anthropic wrapper: generate() and generate_structured()

E. Stage 1: Intake (6 files)

#	File	Description
20	src/stage1/classifier.py	RoBERTa multi-label classifier: train(), predict(), needs_hitl(), save/load
21	src/stage1/aspect_sentiment.py	LLM-based aspect-sentiment extraction: analyze(), analyze_batch()
22	src/stage1/entity_extractor.py	Hybrid regex + LLM: devices, OS, versions, screens, features
23	src/stage1/hitl_checkpoint.py	Confidence flagging, correction recording, retraining data export
24	src/stage1/pipeline.py	Orchestrator: process() and process_with_hitl()
25	src/stage1/__init__.py	Module exports

F. Stage 2: KG + Clustering (6 files)

#	File	Description
26	src/stage2/kg_builder.py	NetworkX DiGraph: add_reviews(), aspect nodes, pagerank, export/load
27	src/stage2/clustering.py	Two-level HDBSCAN: Level 1 by aspect, Level 2 sub-clustering
28	src/stage2/schema_mapper.py	Maps to IssueCluster: infer type, select reps, merge entities, temporal
29	src/stage2/priority_ranker.py	Weighted: PageRank + review_count + sentiment + recency
30	src/stage2/pipeline.py	Orchestrator: KG -> Clustering -> Schema -> Ranking
31	src/stage2/__init__.py	Module exports

G. Stage 3: Translation -- Core Contribution (5 files)

#	File	Description
32	src/stage3/taxonomy.py	5 templates: Zimmermann bug, user story, ISO 25010, Nielsen, compat matrix
33	src/stage3/translator.py	LLM IssueCluster -> IssueSpec: translate(), prompt building, parsing
34	src/stage3/hitl_checkpoint.py	5-dim rubric, threshold check, weak dims, regeneration with feedback
35	src/stage3/pipeline.py	Orchestrator: process() and process_with_hitl() with retry loop
36	src/stage3/__init__.py	Module exports

H. Stage 4b: Response Generation (5 files)

#	File	Description
37	src/stage4b/rag_retriever.py	ChromaDB: 5 sources, index_source(), retrieve() with top-k
38	src/stage4b/response_generator.py	Issue-spec-aware: builds context from RAG + spec, generates response
39	src/stage4b/self_refiner.py	Self-critique: specificity/compliance/empathy, revises 2-3 times
40	src/stage4b/pipeline.py	Orchestrator: ablation toggles for RAG, spec, refinement
41	src/stage4b/__init__.py	Module exports

I. Stage 5: RLHF (7 files)

#	File	Description
42	src/stage5/feedback_collector.py	Dual-stream: quality (5 dims) + compliance (4 flags), KTO/DPO/PPO export
43	src/stage5/kto_trainer.py	KTO with LoRA via TRL: binary good/bad training
44	src/stage5/dpo_trainer.py	DPO with LoRA via TRL: paired preference training
45	src/stage5/constrained_ppo.py	Constrained PPO (CMDP): dual reward models, constrained optimization
46	src/stage5/feedback_propagator.py	Backward: corrections -> Stage 1, rubric -> Stage 3, quality -> Stage 4b
47	src/stage5/pipeline.py	Auto-selects KTO/DPO/PPO by data volume
48	src/stage5/__init__.py	Module exports

J. Evaluation Module (6 files)

#	File	Description
49	src/evaluation/metrics.py	BLEU, ROUGE-L, BERTScore, completeness ratio, Krippendorff alpha
50	src/evaluation/statistical_tests.py	Wilcoxon, Friedman, Nemenyi, Bradley-Terry, McNemar, Bonferroni
51	src/evaluation/experiment1.py	Exp 1: 4 conditions, rubric eval, Wilcoxon comparison, report
52	src/evaluation/experiment2.py	Exp 2: 4 conditions, auto + human metrics, Friedman + Nemenyi
53	src/evaluation/experiment3.py	Exp 3: 3 RLHF variants, Bradley-Terry + McNemar, iteration tracking
54	src/evaluation/ablations.py	A1-A7 runner: full vs ablated comparison, report table

K. API (5 files)

#	File	Description
55	api/main.py	FastAPI app: health check, 4 route groups
56	api/routes/intake.py	POST /reviews/intake -- process through Stage 1
57	api/routes/issues.py	GET /issues -- list/retrieve generated issue specs
58	api/routes/responses.py	POST /responses/generate -- generate response for a review
59	api/routes/feedback.py	POST /feedback/quality + /feedback/compliance

L. Scripts, Tests, Data (7 files)

#	File	Description
60	scripts/run_pipeline.py	End-to-end: Stage 1 -> 2 -> 3 -> 4b with JSON output
61	scripts/run_experiment.py	CLI: python3 run_experiment.py [1 2 3 ablations]
62	scripts/download_datasets.py	Downloads MAALEJ, RRGEn, GUZMAN + generates synthetic data
63	tests/test_schemas.py	All Pydantic models: ReviewObject, RubricScores, IssueSpec, etc.
64	tests/test_stage2/test_kg_builder.py	KG: add_reviews, aspects, pagerank, export/load
65	tests/test_evaluation/test_metrics.py	Completeness, rubric agg, Wilcoxon, Friedman, McNemar, B-T
66	data/raw/sample_reviews.json	500 synthetic reviews (balanced across 6 categories)

M. Downloaded Datasets (ALL VERIFIED)

#	Dataset	Details
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67	data/raw/maalej/maalej_reviews.json	50,000 reviews from 189 apps, 12.3 MB, all ratings 1-5
68	data/raw/rrgen/rrgen_reviews.json	310,031 review-response pairs from 58 apps, 144 MB
69	data/raw/rrgen/data/ (23 files)	Train 280K + Valid 15K + Test 16K + embeddings, 1.7 GB total
70	data/raw/guzman/guzman_reviews.json	2,062 sentences, 1,040 aspect-sentiment pairs from 8 apps
71	data/raw/guzman/Ground_truth.xlsx	Source xlsx with annotated reviews (319 KB)
72	data/processed/training_data.json	Training-ready format: 500 texts + labels + ratings

Completed Summary

Category	Files	Status
Documentation	6	DONE
Project Setup	3	DONE
Configuration	7	DONE
Common Module	3	DONE
Stage 1: Intake	6	DONE
Stage 2: KG + Clustering	6	DONE
Stage 3: Translation	5	DONE
Stage 4b: Response Gen	5	DONE
Stage 5: RLHF	7	DONE
Evaluation Module	6	DONE
API Routes	5	DONE
Scripts	3 (pipeline + experiment + download)	DONE
Tests	3	DONE
Sample + Synthetic Data	2	DONE
Downloaded Datasets	6 (MAALEJ + RRGGen + GUZMAN)	DONE - ALL VERIFIED
Init files	17	DONE
TOTAL	78 content + 17 init = 95	ALL DONE

RECENTLY COMPLETED (Previously Incomplete)

1. Dataset Download Script [DONE]

File: scripts/download_datasets.py. Downloads MAALEJ, RRGGen, GUZMAN datasets + generates synthetic data. Supports individual or batch download. All 3 datasets verified and processed into JSON.

2. MAALEJ Dataset Download [DONE]

50,000 app reviews from 189 apps downloaded from sealuzh/user_quality GitHub repo. 12.3 MB JSON. Ratings 1-5, avg 106 chars/review. Zero empty entries. Integrity: PASS.

3. RRGGen Dataset Download [DONE]

310,031 review-response pairs from 58 apps downloaded from Google Drive via gdown. 144 MB JSON + 1.7 GB raw data (train 280K, valid 15K, test 16K + embeddings). Avg review 74 chars, avg response 229 chars. Integrity: PASS.

4. GUZMAN Dataset Download [DONE]

2,062 annotated sentences from 8 apps (Spotify, Twitter, WhatsApp, etc.) downloaded from Dabrowski IS-22 repo (same methodology as Guzman & Maalej 2014). 1,040 aspect-sentiment pairs (292 positive, 638 neutral, 110 negative). 894 unique aspects. Integrity: PASS.

5. Synthetic Sample Data [DONE]

500 balanced synthetic reviews generated across 6 categories: bug 30%, feature 18%, usability 17%, performance 14%, praise 12%, compatibility 10%. Training-ready format also generated.

6. Fine-tune RoBERTa Classifier (Run 1: Auto-Labels) [DONE]

First training run on 9,529 reviews (500 synthetic + 5,000 MAALEJ auto-labeled via keyword heuristics + 4,029 RRGGen auto-labeled). F1 micro: 0.8355, F1 macro: 0.7478. Completed in ~50 min on MPS. Limitation: auto-labeling is noisy and approximate.

7. Fine-tune RoBERTa Classifier (Run 2: Real MAALEJ Labels) [DONE]

Retrained on actual human-annotated MAALEJ data (5,008 reviews from Maalej et al. 2016 paper, downloaded from github.com/mohammadzaeem/classification_of_app_reviews) + 500 synthetic = 5,508 total. Original labels: Problem Discovery->bug_report (864), Feature Request->feature_request (444), User Experience->usability (607), Rating->praise (2389), Information Giving/Seeking->other (704). Synthetic data fills gaps for performance (70) and compatibility (50) categories missing from MAALEJ. F1 macro improved from 0.7478 to 0.7992 (+6.9%). Model saved to models/stage1_classifier/. Training log: TRAINING_LOG.md.

8. MAALEJ Labeled Dataset Downloaded [DONE]

Original Maalej et al. (2016) labeled dataset: app_reviews.csv (5,008 reviews with human annotations) + Maalej_Dataset.xlsx (3,691 reviews) + Pan_Dataset.xlsx (1,390 reviews). Source: github.com/mohammadzaeem/classification_of_app_reviews.

INCOMPLETE ITEMS (12 Remaining)

Datasets downloaded, classifier trained on real MAALEJ labels. RRGGen (310K) not used for training because it has no category labels -- used for RAG in Stage 4b. GUZMAN (2K) not used for training because it has aspect-sentiment annotations, not category labels -- used for aspect sentiment in Stage 1 and KG edges in Stage 2.

1. RAG Index Population [INCOMPLETE]

File needed: scripts/populate_rag.py. Must load 5 RAG sources (RRGGen 310K past responses, changelogs, FAQ, etc.) and index into ChromaDB. Datasets now available. Without this, RAG returns empty. Priority: HIGH. Effort: 1-2 hours. Blocked by: Nothing.

2. Constrained PPO Reward Model Inference [INCOMPLETE]

File: src/stage5/constrained_ppo.py. Methods _score_quality() and _score_compliance() return hardcoded 0.5 instead of actual model inference. PPO training produces meaningless rewards. Priority: MEDIUM. Effort: 2-3 hours. Blocked by: Needs trained reward models from feedback data.

3. DPO Training Text Mapping [INCOMPLETE]

File: src/stage5/feedback_collector.py export_dpo_data() returns response IDs as pairs but DPO trainer needs actual prompt+response texts. The ID-to-text mapping is not wired up. Priority: MEDIUM. Effort: 1 hour. Blocked by: Nothing.

4. Gold-Standard Dataset Construction [INCOMPLETE]

Manual research activity: 200-300 review clusters must be scored by 3+ domain experts writing structured issue specs. This is Contribution 5 in the proposal and primary evaluation artifact for Experiment 1. Priority: HIGH. Effort: Weeks. Blocked by: Pipeline running on real data to produce clusters.

5. Unit Tests -- Stage 1 [INCOMPLETE]

Directory exists: tests/test_stage1/ (only __init__.py). Need: test_classifier.py, test_aspect_sentiment.py, test_entity_extractor.py, test_pipeline.py. Priority: MEDIUM. Effort: 1-2 hours. Blocked by: Nothing.

6. Unit Tests -- Stage 3 [INCOMPLETE]

Directory exists: tests/test_stage3/ (only __init__.py). Need: test_taxonomy.py, test_translator.py, test_hitl_checkpoint.py. Priority: MEDIUM. Effort: 1-2 hours. Blocked by: Nothing.

7. Unit Tests -- Stage 4b [INCOMPLETE]

Directory exists: tests/test_stage4b/ (only __init__.py). Need: test_rag_retriever.py, test_response_generator.py, test_self_refiner.py. Priority: MEDIUM. Effort: 1-2 hours. Blocked by: Nothing.

8. Unit Tests -- Stage 5 [INCOMPLETE]

Directory exists: tests/test_stage5/ (only __init__.py). Need: test_feedback_collector.py, test_feedback_propagator.py. Priority: MEDIUM. Effort: 1-2 hours. Blocked by: Nothing.

9. Jupyter Notebooks [INCOMPLETE]

Directory exists: notebooks/ (empty). Need: 01_data_exploration.ipynb, 02_stage1_training.ipynb, 03_kg_visualization.ipynb, 04_experiment_results.ipynb. Priority: MEDIUM. Effort: 2-3 hours.

10. Dockerfile [INCOMPLETE]

Files needed: Dockerfile + docker-compose.yaml. Containerize the API for deployment. Priority: LOW. Effort: 1-2 hours. Blocked by: Nothing.

11. CI/CD Pipeline [INCOMPLETE]

File needed: .github/workflows/test.yml. Run tests + ruff linting on every push. Priority: LOW. Effort: 1 hour. Blocked by: Nothing.

12. Frontend Dashboard for HITL + Multi-language Support [INCOMPLETE]

Web UI (Streamlit/React) for HITL checkpoints + XLM-RoBERTa for multilingual reviews. Priority: LOW. Effort: Days. Blocked by: Nothing.

Recommended Priority Order

Priority	Item	Why First	Effort	Blocked By
1	RAG index population (#1)	Stage 4b generic without RAG	1-2 hrs	Nothing
2	Fix DPO text mapping (#3)	Quick fix, unblocks RLHF	1 hr	Nothing
3	Fix PPO reward placeholders (#2)	Unblocks Constrained PPO	2-3 hrs	Feedback data
4	Unit tests (#5-8)	Catch bugs before experiments	4-6 hrs	Nothing
5	Jupyter notebooks (#9)	Needed for exploration + paper	2-3 hrs	Nothing
6	Gold-standard dataset (#4)	Contribution 5, needed for Exp 1	Weeks	Real clusters
7-12	Docker, CI/CD, frontend, multilingual	Nice-to-have, not blocking	Days	Nothing

Final Summary

Metric	Count
Total files created	97+ (source + configs + data + model)
Source code files	45 (non-init Python files)
Config files	7
Documentation files	7 (+ TRAINING_LOG.md)
Downloaded datasets	3 (MAALEJ 50K + RRGGen 310K + GUZMAN 2K)
MAALEJ labeled dataset	5,008 human-annotated reviews (actual paper data)
Total data points available	362,593
Trained model	RoBERTa classifier, F1 macro 0.7992, 476 MB
Test files	3 (+ 4 more needed)
Originally incomplete	15
Completed so far	8 (datasets + classifier training x2 + labeled data)
Currently incomplete	12
High priority remaining	1 (RAG index)
Medium priority remaining	7

Low priority remaining	3
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